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Towards a Definition of Physiologic Vulnerability in Pediatric Spine Surgery: Identification of Key Risk Factors in a Cohort Study of Children With Neuromuscular Disease Undergoing Spinal Fusion

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Abstract

Study Design: Retrospective cohort study.

Objectives: Preoperative risk stratification using frailty is common for adults but difficult to apply to pediatric populations. We aimed to identify risk factors indicating physiologic vulnerability and predict perioperative complications in children with neuromuscular scoliosis (NMS) and to create a prediction model for physiological vulnerability (PV-5).

Methods: Patients with NMS were identified from the American College of Surgeons National Surgical Quality Improvement Program Pediatric database. The 9442 patients identified were randomly divided into training and testing cohorts. Univariate and multivariable logistic regression were performed; variables significantly associated with complications were evaluated using the Akaike information criterion and area under the curve. Significant variables received weighted scores, and a patient-specific prediction model was generated and evaluated using the Brier score.

Results: Patients with central nervous system abnormality (OR 1.32 [95%CI 1.13-1.53]), hematologic disorder (OR 1.40 [1.06-1.85]), congenital malformation (OR 1.30 [1.1-1.54]), nutritional support (OR 2.21 [1.91-2.57]), and preoperative wound infection (OR 2.3 [1.4-3.76]) were more likely to develop complications after spinal fusion surgery. PV-5 scores were calculated from these risk factors to generate a prediction model. PV-5 scores of 1 (OR: 2.0 [1.27-3.43], $P < 0.004$), 2 (OR: 2.75 [1.63-4.64], $P < 0.001$), 3 (OR: 3.67 [2.18-6.19], $P < 0.001$), 4 (OR: 4.09 [2.39-6.99], $P < 0.001$), and 5+ (OR: 3.58 [1.35-9.47], $P = 0.01$) predicted greater complication risk than PV-5 of zero (accuracy = 89.65%, Brier score = 0.09).

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Conclusions: Using factors associated with complications in children with NMS undergoing spinal fusion surgery, we created a prediction model to illustrate physiologic vulnerability and morbidity. Our model serves as a foundation for further body system-specific investigation.

Keywords
neuromuscular scoliosis, frailty, spinal deformity, spine surgery, deformity correction, physiologic vulnerability, risk stratification, screening tool, pediatric

Introduction

Scoliosis is common in children with neuromuscular disorders or severe neurological impairments.[1] Indications for surgery to address neuromuscular scoliosis (NMS) vary by deformity, severity, and underlying condition. These cases are complex, in part because of comorbid conditions that may increase the likelihood of intraoperative or postoperative complications.[2-4] The benefits of surgery can be significant, including preserved upright posture, improved pulmonary function, reduced aspiration, and overall better health-related quality of life.[5] Patients may choose to pursue surgical intervention based on their understanding of preoperative assessment, risk stratification, and perceived benefit.

Recently, physiologic vulnerability, often conflated with frailty, has become a topic of interest in spine surgery, including in adult patients undergoing surgery for spinal deformity correction.[6-8] A common thread is that aging triggers multisystem body dysregulation and greater susceptibility to adverse health outcomes, known as frailty.[9,10] The Fried frailty phenotype was developed to standardize the definition of frailty in the geriatric population and is based on 5 domains: unintentional decrease in body weight, fatigue, decreased grip strength, slow walking, and decline in physical activity.[10] Fried's phenotype was described in the geriatric population, but the concept of physiologic vulnerability may not be age-specific. An attempt has been made to operationalize the term "frailty" to emphasize clinical rather than physiologic phenotypes.[10] Frailty scores, such as the 5- and 11-factor modified frailty indexes (mFI-5 and mFI-11), are based on clinical characteristics that have been shown to increase a patient's risk for poor outcomes. However, it is difficult to extrapolate these indices outside their intended populations, and these brief indices may combine disease-related factors with age-related factors, which distinguish the Fried phenotype.[11-13]

Children with NMS harbor many of the traits traditionally considered reflective of the frailty clinical phenotype, including low body weight and osteosarcopenia; additionally, children with neuromuscular disease often have comorbidity profiles and conditions that would suggest "frailty."[11-13] To date, no study has aimed to operationalize physiologic vulnerability in the population of pediatric patients with NMS. To do so, we sought a measure of physiologic vulnerability independent of age and age-related decline in health and function.

In this study, we performed an exploratory analysis to identify factors associated with greater risk of complications to implicate key body systems associated with physiologic vulnerability in children with NMS. We then developed a preoperative risk calculator to predict the incidence of complications after pediatric deformity correction for NMS. We hypothesized that there is a specific set of factors that identify physiologically vulnerable patients at risk for complications in this specific population, which differs from adults, and can be used to predict morbidity after spinal fusion surgery.

Methods

Data Source

We performed a retrospective cohort study using the American College of Surgeons National Surgical Quality Improvement Program Pediatric (NSQIP-P) database (https://www.facs.org/quality-programs/data-and-registries/pediatric/). NSQIP-P tracks and audits 30-day perioperative outcomes (perioperative and surgical) for individuals ≤18 years. Variables are collected based on a systematic sampling process that allows for proportional diversity in selection by using an 8-day sampling cycle from more than 60 pediatric institutions. The NSQIP-P data are audited and validated continually and organized into participant user files for quality improvement and outcomes research. Because NSQIP-P participant user file datasets do not have patient-identifying information, the study was exempt from ethics review as determined by the local institutional review board and human subjects protection office.

Patient Population

NSQIP-P participant user files from 2012 to 2020 were queried. Codes from the International Classification of Diseases (ICD)-9 and −10 were used to identify patients with NMS. All relevant Current Procedural Terminology (CPT) codes encoding arthrodesis for spinal deformity (eg, scoliosis, kyphosis) up to 13 or more vertebral segments, including pelvic fixation, were used to screen patients undergoing spinal fusion (Table 1). Patients with data missing for any outcomes evaluated and variables with less than 70% data were excluded.

Table 1. ICD-9, ICD-10, and CPT Codes Used in the Study With Their Definitions.

ICD-9 Code	Definition
334	Spinocerebellar disease
335	Anterior horn cell disease
335.1	Spinal muscular atrophy
343.8	Other specified infantile cerebral palsy
343.9	Infantile cerebral palsy, unspecified
359.1	Hereditary progressive muscular dystrophy
741	Spina bifida with hydrocephalus
741.9	Spina bifida without mention of hydrocephalus
756.1	Congenital anomalies of spine
767.4	Injury to spine and spinal cord due to birth trauma
806	Closed fracture of cervical vertebra with spinal cord injury
907.2	Late effects of injuries to the nervous system
952	Spinal cord injury without evidence of spinal bone injury
8102	Other cervical fusion of the anterior column, anterior technique
8103	Other cervical fusion of the posterior column, posterior technique
8104	Dorsal and dorsolumbar fusion of the anterior column, anterior technique
8105	Dorsal and dorsolumbar fusion of the posterior column, posterior technique
8106	Lumbar and lumbosacral fusion of the anterior column, anterior technique
8107	Lumbar and lumbosacral fusion of the posterior column, lateral transverse process technique
8108	Lumbar and lumbosacral fusion of the anterior column, posterior technique
ICD-10 Code	**Definition**
G80.9	Cerebral palsy, unspecified
Q05.0	Cervical spina bifida with hydrocephalus
Q05.1	Thoracic spina bifida with hydrocephalus
Q05.2	Lumbar spina bifida with hydrocephalus
Q05.3	Sacral spina bifida with hydrocephalus
Q05.4	Unspecified spina bifida with hydrocephalus
Q05.5	Cervical spina bifida without hydrocephalus
Q05.6	Thoracic spina bifida without hydrocephalus
Q05.7	Lumbar spina bifida without hydrocephalus
Q05.8	Sacral spina bifida without hydrocephalus
Q05.9	Spina bifida, unspecified
G12.0	Infantile spinal muscular atrophy, type I [Werdnig-Hoffman]
G12.1	Other inherited spinal muscular atrophy
G12.8	Other spinal muscular atrophies and related syndromes
G12.9	Spinal muscular atrophy, unspecified
G11.1	Early-onset cerebellar ataxia
G71.0	Muscular dystrophy
S13.10	Subluxation and dislocation of unspecified cervical vertebrae

(continued)

Table 1. (continued)

ICD-10 Code	Definition
S13.11	Subluxation and dislocation of C0/C1 cervical vertebrae
S13.12	Subluxation and dislocation of C1/C2 cervical vertebrae
S13.13	Subluxation and dislocation of C2/C3 cervical vertebrae
S13.14	Subluxation and dislocation of C3/C4 cervical vertebrae
S13.15	Subluxation and dislocation of C4/C5 cervical vertebrae
S13.16	Subluxation and dislocation of C5/C6 cervical vertebrae
S13.17	Subluxation and dislocation of C6/C7 cervical vertebrae
S13.18	Subluxation and dislocation of C7/T1 cervical vertebrae
S22.00	Fracture of unspecified thoracic vertebra
S22.01	Fracture of first thoracic vertebra
S22.02	Fracture of second thoracic vertebra
S22.03	Fracture of third thoracic vertebra
S22.04	Fracture of fourth thoracic vertebra
S22.05	Fracture of T5-T6 vertebra
S22.06	Fracture of T7-T8 vertebra
S22.07	Fracture of T9-T10 vertebra
S22.08	Fracture of T11-T12 vertebra
S33.10	Subluxation and dislocation of unspecified lumbar vertebra
S33.11	Subluxation and dislocation of L1/L2 lumbar vertebra
S33.12	Subluxation and dislocation of L2/L3 lumbar vertebra
S33.13	Subluxation and dislocation of L3/L4 lumbar vertebra
S33.14	Subluxation and dislocation of L4/L5 lumbar vertebra
G95.9	Disease of spinal cord, unspecified
CPT Code	**Definition**
22800	Arthrodesis, posterior, for spinal deformity, with or without cast; up to 6 vertebral segments
22802	Arthrodesis, posterior, for spinal deformity, with or without cast; 7 to 12 vertebral segments.
22804	Arthrodesis, posterior, for spinal deformity, with or without cast; 13 or more vertebral segments
22808	Arthrodesis, anterior, for spinal deformity, with or without cast; 2 to 3 vertebral segments
22810	Arthrodesis, anterior, for spinal deformity, with or without cast; 4 to 7 vertebral segments
22812	Arthrodesis, anterior, for spinal deformity, with or without cast; 8 or more vertebral segments
22818	Kyphectomy, circumferential exposure of spine and resection of vertebral segment(s) (including body and posterior elements), 1-2 segments
22819	Kyphectomy, circumferential exposure of spine and resection of vertebral segment(s) (including body and posterior elements), 3 or more segments
22848	Pelvic fixation (attachment of caudal end of instrumentation to pelvic Bony structures) other than sacrum

ICD, International Classification of Diseases; CPT, Current Procedural Terminology.

Data Analysis and Prediction Modeling

All variables in the NSQIP-P database were evaluated by 4 independent reviewers for relevance. Patient demographic

factors (age, sex, race) and 44 preoperative variables were selected in advance for evaluation. The primary study outcome was the incidence of any postoperative complication within 30 days of surgery, which was defined as a composite of the following outcome variables: urinary tract infection, pneumonia, vein thrombosis requiring therapy, superficial incisional surgical site infection (SSI), deep incisional SSI, organ/space SSI, deep wound disruption/dehiscence, unplanned intubation, progressive renal insufficiency, acute renal failure, coma >24 hours, cerebrovascular accident/stroke or intracranial hemorrhage, seizure disorder, nerve injury, cardiac arrest requiring cardiopulmonary resuscitation, *Clostridium difficile* colitis, sepsis, septic shock, and central line-associated bloodstream infection. Categorical or dichotomous variables were compared using the Chi-squared test of independence.

A training cohort comprising 70% of the overall cohort was randomly selected. Univariate analysis was performed to identify which variables were associated with complications (P-value <0.10). An Akaike information criterion (AIC) analysis was then used to estimate the relative predictive quality. Additionally, an area under the curve (AUC) analysis was performed to identify the combination of variables predicting the primary outcome. Subsequent multivariable analysis was performed to identify independent associations with the outcome of complications after controlling for confounding variables. A high-throughput approach involving multiple iterations of logistic regression analyses was used to find the best model. The model with a combination of variables that produced the lowest AIC value and highest AUC score was used to create our final patient-specific prediction model. A variance inflation factor (VIF) was calculated for each of the variables in the final prediction model to assess multicollinearity between features. A VIF value >5 was considered as contributing considerably to multicollinearity.

The remaining 30% of patients in the cohort were allocated to a testing/validation group (ie, a "naïve" patient population used to validate our final prediction model). The model with a combination of variables that produced the lowest AIC value/highest AUC score was used. Odds ratios (ORs) were converted to integers and used to assign a weighted score for each variable—a score of 1 for an OR of 1-2 and a score of 2 for OR >2. ORs were converted to probabilities and used to create a prediction scoring model termed the 5-factor physiologic vulnerability score (PV-5) to predict complications.[14,15] A score greater than the reference score of zero indicated a higher risk of postoperative complications. This scale was applied to the validation cohort. Subsequent analysis was performed to evaluate our model's predictive performance as a risk stratification model.

To evaluate our model's diagnostic performance, a confusion matrix was generated to compare true positives with predicted positives. Accuracy, sensitivity, and specificity were calculated at various pediatric physiological vulnerability scores to determine the best score cutoff. This analysis workflow was performed in an all-encompassing patient age group and in 2 separate age groups (0-9 and ≥10 years old). To validate our model's predictive capabilities, all possible combinations of predictors were analyzed to create a risk calculator that provided a probabilistic output. Our model's predictive performance was evaluated using a Brier score to generate an accuracy and Brier score loss measurement.[16]

Statistical analysis was performed with the Sci-Kit Learn machine learning package in Python (version 1.1.2, Python Software Foundation, https://www.python.org/).

Results

There were 9442 patients with NMD who underwent posterior spinal fusion for spinal deformity identified. Of this total, 6609 and 2833 patients were allocated to the training and validation cohorts, respectively (Figure 1). Patient baseline demographic, clinical, and surgical characteristics for the entire study cohort are reported in Table 2. Patient characteristics were further subdivided to include the percentages within the complication and non-complication cohorts of patients.

The median age of patients in the study (across both cohorts) was 13 years (11-15). Most children were ≥10 years old, but 12.67% of patients were <10 years old (Table 2). Overall, 46.45% were male and 53.65% were female. Most patients identified as White (67.59%), followed by Black (13.93%), Asian (3.36%), American Indian or Alaskan Native (0.57%), Native Hawaiian/Pacific Islander (0.42%), and other races (0.56%), with 13.75% unknown. Most patients were underweight (54.67%), 33.29% were within normal range, 11.23% were overweight, and 0.82% were obese. Many patients underwent pelvic fixation procedures (2958 [30.27%]), 346 (11.70%) of which were performed in the 0-9 age group and 2612 (88.3%) in the 10+ age group.

Reported preoperative medical comorbidities included premature birth (28.46%), impaired cognitive status at the time of surgery (66.50%), ostomy (31.82%), nutritional support at the time of surgery (32.02%), oxygen support at the time of operation (6.93%), gastrointestinal disease (28.52%), ventilator dependence (9.53%), history of asthma (15.56%), and history of bronchopulmonary disorder (16.34%) (Table 2). Previous neurological conditions included seizure disorder (36.49%), cerebral palsy (37.50%), and structural central nervous system (CNS) abnormality (34.79%). Congenital malformations were reported in 67.09% of patients, along with hematologic disorder (5.21%) and history of wound infection (1.18%). Most patients (83.05%) had no cardiac risk factors, but 8.48% had minor, 7.83% had major, and 0.64% had severe cardiac risk factors. Most patients (79.29%) had an American Society of Anesthesiologists (ASA) classification ≥III, 17.67% were ASA II, 2.99% were ASA I, and 0.05% had no ASA score assigned.

The primary outcome of at least 1 postoperative complication was present in 1132 (11.99%) patients (Table 3). The most common complications were pneumonia (24.91%), unplanned intubation (18.73%), deep incisional SSI (16.70%), urinary tract infection (16.70%), sepsis (16.17%), and superficial SSI (14.93%) (Table 4). The least reported

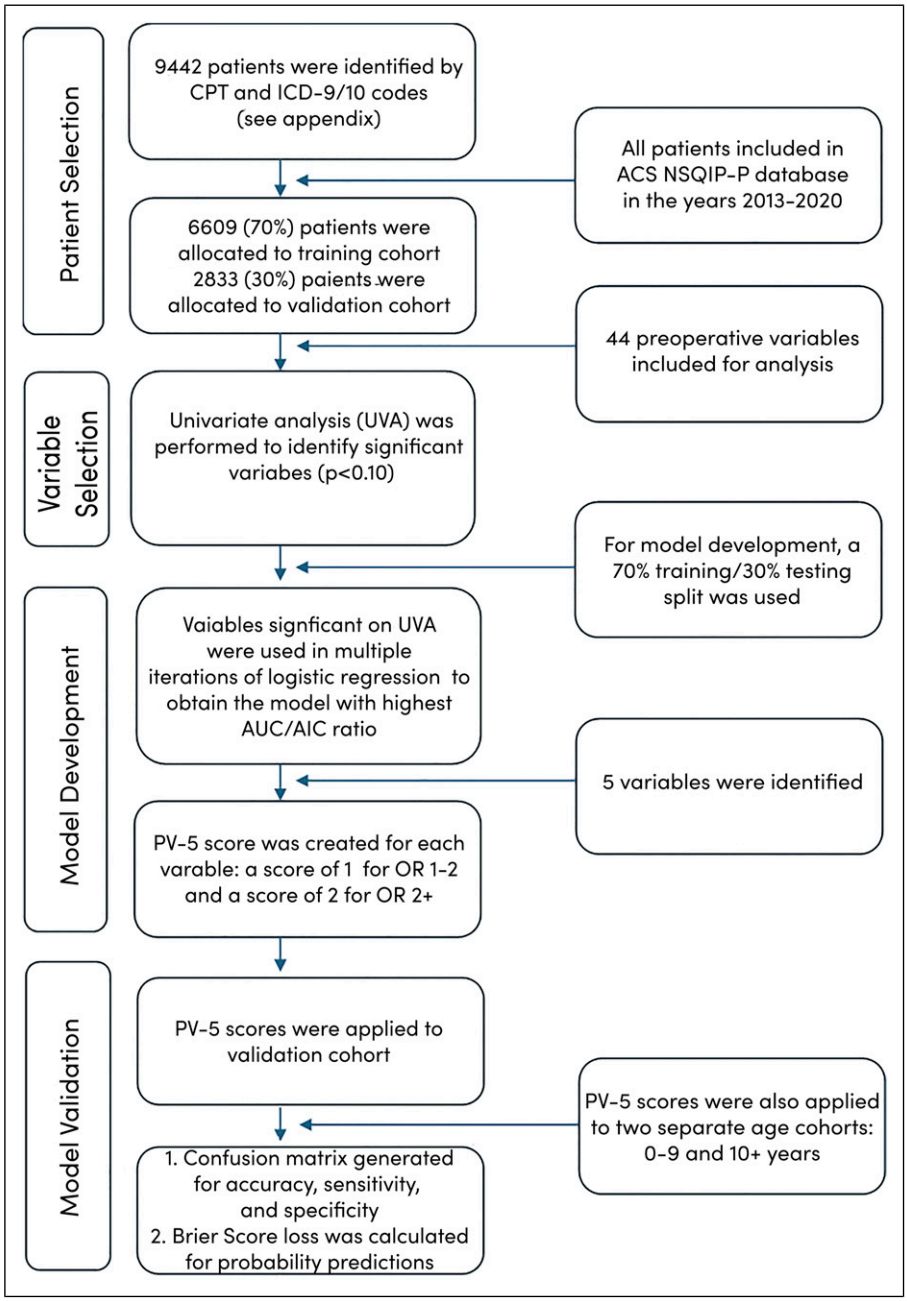

Figure 1. Diagram of analysis workflow.
Note. CPT, Common Procedural Terminology; ICD, International Classification of Diseases; ACS-NSQIP-P, American College of Surgeons National Surgical Quality Improvement Program–Pediatric; UVA, univariate analysis, AUC, area under the curve; AIC, Akaike information criterion; PV-5, 5-factor physiological vulnerability score; OR, odds ratio.

complications were coma >24 hours (0.27%), stroke or intracranial hemorrhage (0.27%), acute renal failure (0.53%), and central line–associated bloodstream infection (0.53%). The median total operation time was 308 (237-395) minutes, and the median length of hospital stay was 5 days (4-7). A small percentage of patients (164 [1.74%]) were hospitalized for >30 days (Table 3). Some patients (749 [7.93%]) required new ventilation support; 81 (0.86%) required ventilation for >30 days, and the mean number of days on mechanical ventilation was 2.28 ± 6.69. Some patients had non-home discharges (589 [6.24%]), but this may include patients who reside in a facility or hospital who were not admitted from "home"; 788 (8.35%) were readmitted within 30 days, and 30 (0.32%) died within 30 days.

Univariate Analysis

Black patients were significantly more likely to develop complications compared with White patients (OR = 1.21, $P = 0.04$)

Table 2. Baseline Demographic, Clinical, and Surgical Variables for Children With Neuromuscular Disease Undergoing Spinal Deformity Correction (n = 9442).

Variable	Value	Non-complication Cohort (n = 8310)	Complication Cohort (n = 1132)	Univariate (95% CI)	*P*-value
Median age (years)	13 (11-15)	13 (11-15)	13 (11-15)	0.99 (0.97-1.01)	0.43
Age range (years)					
0-9	1196 (12.67%)	1070	126	Ref	
10+	8246 (87.33%)	7240	1006	1.18 (0.97-1.44)	0.11
Sex					
Male	4376 (46.45%)	3861 (88.23%)	515 (10.17%)	Ref	
Female	5066 (53.65%)	4449 (87.82%)	617 (12.18%)	0.96 (0.85-1.09)	0.34
Race					
White	6382 (67.59%)	5661 (88.70%)	721 (11.30%)	Ref	
Black	1315 (13.93%)	1140 (86.69%)	175 (13.31%)	1.21 (1.01-1.44)	0.04
Asian	317 (3.36%)	281 (88.64%)	36 (11.36%)	1.01 (0.97-1.44)	0.97
American Indian or Alaskan native	54 (0.57%)	49 (90.74%)	5 (9.26%)	0.80 (1.01-1.44)	0.64
Native Hawaiian/Pacific Islander	23 (0.42%)	19 (82.61%)	4 (17.39%)	1.65 (0.46-0.56)	0.46
Other	53 (0.56%)	42 (79.25%)	11 (20.75%)	2.12 (1.05-4.29)	0.04
Unknown	1298 (13.75%)	1118 (86.13%)	180 (13.87%)	1.26 (1.06-1.44)	0.01
BMI					
Underweight	5162 (54.67%)	4529 (87.74%)	633 (12.26%)	1.09 (0.95-1.25)	0.22
Normal	3143 (33.29%)	2786 (88.64%)	357 (11.36%)	Ref	
Overweight	1060 (11.23%)	931 (87.83%)	129 (12.17%)	1.08 (0.87-1.34)	0.48
Obese	77 (0.82%)	64 (83.12%)	13 (16.88%)	1.59 (0.86-2.91)	0.14
Premature birth	2687 (28.46%)	2335 (86.90%)	352 (13.10%)	1.15 (1.01-1.32)	**0.04**
Impaired cognitive status at time of surgery	6279 (66.50%)	5334 (84.95%)	945 (15.05%)	2.82 (2.40-3.32)	**<0.001**
Nutritional support at time of surgery	3023 (32.02%)	2458 (81.31%)	565 (18.69%)	2.37 (2.09-2.69)	**<0.001**
Oxygen support at time of operation	654 (6.93%)	528 (80.73%)	126 (19.27%)	1.85 (1.50-2.27)	**<0.001**
Gastrointestinal disease	2693 (28.52%)	2224 (82.58%)	469 (17.42%)	1.94 (1.70-2.20)	**<0.001**
Ventilator dependence	900 (9.53%)	760 (84.44%)	140 (15.56%)	1.40 (1.16-1.70)	**<0.001**
History of asthma	1469 (15.56%)	1222 (83.19%)	247 (16.81%)	1.62 (1.39-1.89)	**<0.001**
History of bronchopulmonary disorder	1543 (16.34%)	1282 (83.08%)	261 (16.92%)	1.64 (1.41-1.91)	**<0.001**
Cardiac risk factors	1440 (16.95%)	1383 (96.04%)	217 (3.96%)	1.19 (1.01-1.39)	**0.03**
Seizure disorder	3445 (36.49%)	2830 (82.15%)	615 (17.85%)	2.30 (2.03-2.61)	**<0.001**
Cerebral palsy	3541 (37.50%)	2965 (83.73%)	576 (16.27%)	1.87 (1.65-2.12)	**<0.001**
Structural CNS abnormality	3285 (34.79%)	2788 (84.87%)	497 (15.13%)	1.55 (1.37-1.76	**<0.001**
Congenital malformation	6335 (67.09%)	5497 (86.77%)	838 (13.23%)	1.46 (1.27-1.68)	**<0.001**
Hematologic disorder	492 (5.21%)	398 (80.89%)	94 (19.11%)	1.80 (1.42-2.27)	**<0.001**
Preoperative wound infection	111 (1.18%)	80 (72.07%)	31 (28.93%)	2.90 (1.90-4.41)	**<0.001**
Preoperative steroid use	256 (2.71%)	220 (85.94%)	36 (14.06%)	1.21 (0.84-1.73)	0.3
Ostomy	3004 (31.82%)	2450 (81.56%)	554 (18.44%)	2.69 (2.32-3.10)	**<0.001**
CPR within 7 days of surgery	5 (0.05%)	5 (100.00%)	0 (0.00%)	1.84 (0.21-16.44)	0.59
Childhood malignancy					
Current cancer or active treatment	35 (0.37%)	33 (94.29%)	2 (5.71%)	0.44 (0.11-1.84)	0.26
History of cancer	140 (1.48%)	130 (92.86%)	10 (7.14%)	0.56 (0.29-1.07)	0.08
Inotropic support at time of operation	92 (0.97%)	81 (88.04%)	11 (11.96%)	1.00 (0.53-1.88)	0.99
ASA status[a]					
I	282 (2.99%)	277 (98.23%)	5 (1.77%)	Ref	
II	1668 (17.67%)	1588 (95.20%)	80 (4.80%)	2.79 (1.12-6.95)	**0.03**
≥III	7487 (79.29%)	6442 (86.04%)	1045 (13.96%)	8.99 (3.70-21.81)	**<0.001**
None assigned	5 (0.05%)	3 (60.00%)	2 (40.00%)	36.93 (5.02-271.77)	**<0.001**
Procedure[b]					
CPT 22800	1271 (13.46%)	1151 (90.56%)	120 (9.44%)	Ref	
CPT 22802	1988 (21.05%)	1869 (94.01%)	119 (5.99%)	0.61 (0.47-0.80)	**<0.001**

(continued)

Table 2. (continued)

Variable	Value	Non-complication Cohort (n = 8310)	Complication Cohort (n = 1132)	Univariate (95% CI)	*P*-value
CPT 22804	5950 (63.02%)	5092 (85.58%)	858 (14.42%)	1.62 (2.32-1.98)	**<0.001**
CPT 22808	39 (0.41%)	31 (79.49%)	8 (20.51%)	2.48 (1.12-5.51)	**0.03**
CPT 22810	137 (1.45%)	114 (83.21%)	23 (16.79%)	1.94 (1.19-3.15)	**0.008**
CPT 22812	57 (0.60%)	53 (92.98%)	4 (7.02%)	0.72 (0.26-2.03)	0.54
CPT 22848	2958 (30.27%)	2485 (84.01%)	473 (15.99%)	1.68 (1.48-1.91)	**<0.001**
Ages 0-9	346 (11.70%)	—	—	—	—
Ages 10+	2612 (88.3%)	—	—	—	—
Median total operation time (min)	308 (237-395)	305 (235-391)	335 (256-428)	1.00 (1.00-1.00)	**<0.001**

Values reported as number (%) unless otherwise indicated.
P values in boldface type are significant at $P < 0.05$.
[a]Five without ASA classification assigned.
[b]See Table 1.
BMI, body mass index; CNS, central nervous system; ASA, American Society of Anesthesiologists; CPR, cardiopulmonary resuscitation; CPT, Common Procedural Terminology.

Table 3. Postoperative Outcomes for Children With Neuromuscular Scoliosis Undergoing Spinal Deformity Correction (n = 9442).

Outcome	Value
Postoperative complications	1132 (11.99%)
Median length of hospital stay in days (range)	5 (4-7)
Hospitalization >30 days	164 (1.74%)
New ventilation support required	749 (7.93%)
>30 days on mechanical ventilation	81 (0.86%)
Mean days on mechanical ventilation (±SD)	2.28 ± 6.69
Non-home discharge	589 (6.24%)
Rehabilitation[a]	209 (35.48%)
Skilled nursing facility[a]	128 (21.73%)
Acute care[a]	68 (11.54%)
Unknown[a]	162 (27.50%)
Died[a]	22 (3.74%)
Home discharge	8853 (93.76%)
Readmission within 30 days	788 (8.35%)
Death within 30 days	30 (0.32%)

Values reported as number (%) unless otherwise indicated.
[a]Values reported as a percentage of non-home discharges or death.
SD, standard deviation.

Table 4. Postoperative Complication Rates of Patients Having at Least One Complication (N = 1132).

Postoperative Complication	Frequency
Pneumonia	282 (24.91%)
Unplanned intubation	212 (18.73%)
Urinary tract infection	189 (16.70%)
Deep incisional SSI	189 (16.70%)
Sepsis	183 (16.17%)
Superficial incisional SSI	169 (14.93%)
Deep wound disruption/dehiscence	115 (10.16%)
Organ/space SSI	75 (6.63%)
Septic shock	47 (4.15%)
Cardiac arrest requiring CPR	41 (3.62%)
C. diff colitis	41 (3.62%)
Vein thrombosis requiring therapy	35 (3.09%)
Death	30 (2.65%)
Nerve injury	19 (1.68%)
Seizure disorder	16 (1.41%)
Progressive renal insufficiency	10 (0.88%)
Central line-associated bloodstream infection	6 (0.53%)
Acute renal failure	6 (0.53%)
CVA/stroke or intracranial hemorrhage	3 (0.27%)
Coma >24 hours	3 (0.27%)

SSI, surgical site infection; CPR, cardiopulmonary respiration; *C. diff*, *Clostridium difficile*; CVA, cerebrovascular accident.

(Table 2). Patient age, sex, and body mass index were not significant predictors of complications. Of the 44 preoperative variables studied on univariate analysis, premature birth (P = 0.04), impaired cognitive status at time of surgery ($P < 0.001$), nutritional support at time of surgery ($P < 0.001$), oxygen support at time of operation ($P < 0.001$), gastrointestinal disease ($P < 0.001$), ventilator dependence ($P < 0.001$), history of asthma ($P < 0.001$), history of bronchopulmonary disorder ($P < 0.001$), cardiac risk factors ($P = 0.03$), seizure disorder ($P < 0.001$), cerebral palsy ($P < 0.001$), structural CNS abnormality ($P < 0.001$), congenital malformation ($P < 0.001$), hematologic disorder ($P < 0.001$), preoperative wound infection ($P < 0.001$), ostomy ($P < 0.001$), ASA grade II and ASA grade ≥ III ($P = 0.03$ and $P < 0.05$, respectively), pelvic fixation ($P < 0.001$), and total operating room time ($P < 0.001$) were all associated with the occurrence of complications (Table 2). OR time and pelvic fixation rates portray the variation of surgeries being performed but were not included in the final PV model.

Multivariable Analysis

Multivariable analysis results demonstrated that congenital malformation (OR: 1.30, 95% confidence interval (CI): 1.10-1.54, P = 0.002), structural CNS abnormality (OR 1.32, CI: 1.13-1.53, P <

Table 5. Multivariable Logistic Regression Results of Model With Comorbidity Probabilities as Identified by Area Under the Curve (AUC) and Akaike Information Criteria (AIC) Analysis.

Comorbidity Variable	Odds Ratio (95% CI)	*P*-value	Proposed PV-5 Points	Probability: Without Comorbidity	Probability: With Comorbidity	Variance Inflation Factor
Congenital malformation	1.30 (1.10-1.54)	0.002	+1	9.46%	12.01%	1.68
Structural CNS abnormality	1.32 (1.13-1.53)	<0.001	+1	10.31%	13.18%	1.53
Hematologic disorder	1.40 (1.06-1.85)	0.02	+1	11.60%	15.52%	1.07
Preoperative nutritional support	2.21 (1.91-2.57)	<0.001	+2	8.83%	17.64%	1.43
Preoperative wound infection	2.30 (1.40-3.76)	<0.001	+2	10.19%	20.70%	1.02

CI, confidence interval; PV-5, 5-factor physiologic vulnerability score; CNS, central nervous system.

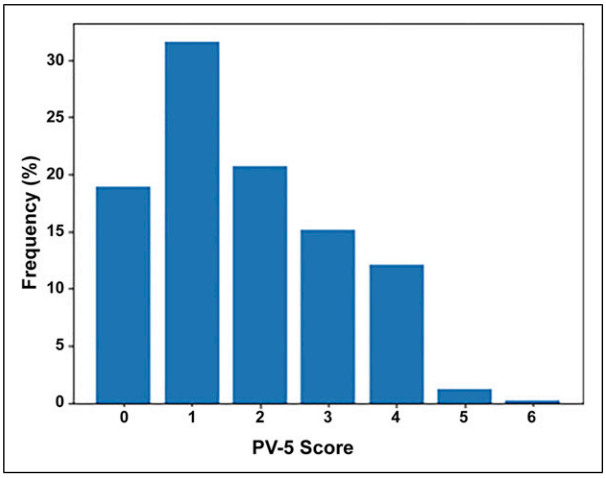

Figure 2. Distribution of 5-factor physiological vulnerability (PV-5) scores within validation cohort.

0.001), hematologic disorder (OR 1.40, CI: 1.06-1.85, $P = 0.02$), preoperative nutritional support (OR: 2.21, CI: 1.91-2.57), $P < 0.001$), and preoperative wound infection (OR 2.30, CI: 1.40-3.76, $P < 0.001$) were all significant predictors of postoperative complications (AIC = 198.434, AUC = 0.77). (Table 5). The VIF results demonstrated minimal multicollinearity among the variables used in the multivariable model (Table 5).

Validation of the PV-5 Score

The validation cohort consisted of 2361 patients with demographic profiles (age, sex, comorbid conditions) like those of the training cohort (data not shown). Most patients in the validation cohort had PV-5 scores of 1 or 2 (Figure 2). Patients of all ages with PV-5 scores of 1 (OR: 2.08, CI: 1.27-3.43, $P < 0.004$), 2 (OR: 2.75, CI: 1.63-4.64, $P < 0.001$), 3 (OR: 3.67, CI: 2.18-6.19, $P < 0.001$), 4 (OR: 4.09, CI: 2.39-6.99, $P < 0.001$), and 5+ (OR: 3.58, CI: 1.35-9.47, $P = 0.01$) were significantly more likely to experience complications than those with a score of zero (Table 6). Patients ≥10 years with PV-5 scores of 1 (OR: 2.23, CI: 1.34-3.71, $P = 0.002$), 2 (OR: 2.71, CI: 1.57-4.67, $P < 0.001$), 3 (OR: 3.95, CI: 2.29-6.81, $P < 0.001$), 4 (OR: 4.27, CI: 2.43-7.51, $P < 0.001$), and 5+ (OR: 3.02, CI: 0.96-9.50, $P = 0.04$) were significantly more likely to experience complications than those with a score of zero (Table 6). Within the 0-9 age group, PV-5 scores failed to significantly predict complication rates.

As PV-5 score increased from zero (low risk) to 4+ (high risk), a stepwise increase was observed in rates of total complications (4.99% to 20.05%), urinary tract infection (0.89% to 3.59%), pneumonia (1.27% to 5.51%), vein thrombosis requiring therapy (0.06% to 0.72%), superficial incisional SSI (0.83% to 2.56%), deep incisional SSI (0.94% to 3.04%), organ/space SSI (0.33% to 1.36%), deep wound disruption/dehiscence (0.39% to 2.16%), unplanned intubation (0.72% to 4.39%), *C. difficile* colitis (0.17% to 1.28%), sepsis (0.66% to 3.51%), septic shock (0.28% to 0.96%), and death (0.17% to 0.80%) (Table 7). PV-5 score cutoffs at 2 or 3 or greater were significantly associated with higher odds of mortality compared with PV-5 scores of zero (Table 8).

Compared with patients without any comorbidities, the probability (OR) of experiencing any complication ranged from 5.29% (OR = 1.3) for patients with congenital malformation alone to 34.41% (OR = 12.21) for patients with all 5 comorbidities (Table 9). Our probabilistic outputs capture all combinations of preoperative risk factors and were subsequently embedded into a surgical risk calculator. Our prediction model demonstrated excellent performance, producing a Brier score loss of 0.0906 (a Brier score of zero indicating perfect accuracy and a Brier score of 1 indicating perfect inaccuracy) and an overall accuracy of 89% (Table 9).

For patients of all ages, the optimal cutoff for PV-5 score to predict the incidence of complications was 3 (accuracy = 0.70, sensitivity = 0.74, specificity = 0.44), with PV-5 scores of 5+ accurately predicting complications in 99% of patients (accuracy = 0.89, sensitivity = 0.99, specificity = 0.03). For patients ages 0-9 and 10+ years, the optimal PV-5 cutoffs were 4 (accuracy = 0.73, sensitivity = 0.79, specificity = 0.37) and 3 (accuracy = 0.72, sensitivity = 0.76, specificity = 0.44), respectively (Table 10).

Discussion

In this study, we have identified a set of variables that suggest physiological vulnerability in children undergoing spinal

Table 6. Univariate Logistic Regression of Proposed PV-5 Scoring System Using the Validation Dataset.

	OR (95% CI), *P*-value		
PV-5 Score	All Ages	Ages 0-9 years	Ages 10+ years
0	[Reference]		
1	2.08 (1.27-3.43), $P < 0.004$	0.96 (0.10-9.10), $P = 0.97$	2.23 (1.34-3.71), $P = 0.002$
2	2.75 (1.63-4.64), $P < 0.001$	3.05 (0.36-25.60), $P = 0.30$	2.71 (1.57-4.67), $P < 0.001$
3	3.67 (2.18-6.19), $P < 0.001$	2.62 (0.31-22.28), $P = 0.38$	3.95 (2.29-6.81), $P < 0.001$
4	4.09 (2.39-6.99), $P < 0.001$	3.40 (0.40-29.04), $P = 0.26$	4.27 (2.43-7.51), $P < 0.001$
5+	3.58 (1.35-9.47), $P = 0.01$	5.71 (0.45-73.19), $P = 0.18$	3.02 (0.96-9.50), $P = 0.04$

PV-5, 5-factor physiologic vulnerability score; OR, odds ratio; CI, confidence interval.

Table 7. Postoperative Complication Rates Stratified by PV-5 Score (N = 9442).

	PV-5 Score					
Complication	0	1	2	3	4+	*P*
Urinary tract infection	0.89%	1.35%	2.37%	2.90%	3.59%	**<0.001**
Pneumonia	1.27%	2.17%	2.37%	5.59%	5.51%	**<0.001**
Vein thrombosis requiring therapy	0.06%	0.26%	0.53%	0.48%	0.72%	**0.02**
Superficial incisional SSI	0.83%	1.51%	2.27%	2.28%	2.56%	**<0.001**
Deep incisional SSI	0.94%	1.88%	2.22%	2.41%	3.04%	**<0.001**
Organ/space SSI	0.33%	0.33%	1.37%	1.10%	1.36%	**<0.001**
Deep wound disruption/dehiscence	0.39%	1.05%	1.58%	1.31%	2.16%	**<0.001**
Unplanned intubation	0.72%	1.88%	2.59%	4.76%	4.39%	**<0.001**
Progressive renal insufficiency	0.00%	0.13%	0.05%	0.21%	0.16%	0.37
Acute renal failure	0.06%	0.00%	0.11%	0.07%	0.16%	0.38
Coma >24 hours	0.00%	0.03%	0.00%	0.00%	0.16%	0.09
CVA/stroke or intracranial hemorrhage	0.00%	0.00%	0.00%	0.07%	0.16%	0.06
Seizure disorder	0.11%	0.13%	0.11%	0.34%	0.24%	0.39
Nerve injury	0.11%	0.26%	0.16%	0.21%	0.24%	0.81
Cardiac arrest requiring CPR	0.39%	0.36%	0.21%	0.76%	0.64%	0.12
C. diff colitis	0.17%	0.16%	0.37%	0.69%	1.28%	**<0.001**
Sepsis	0.66%	1.15%	2.64%	2.90%	3.51%	**<0.001**
Septic shock	0.28%	0.26%	0.47%	0.90%	0.96%	**0.004**
Central line-associated bloodstream infection	0.00%	0.10%	0.16%	0.00%	0.00%	0.19
Death	0.17%	0.16%	0.16%	0.62%	0.80%	**0.001**
All complications	4.99%	9.01%	13.56%	17.93%	20.05%	**<0.001**

P-values in boldface type are significant at $P < 0.05$.
PV-5, 5-factor physiologic vulnerability score; SSI, surgical site infection; CVA, cerebrovascular accident; CPR, cardiopulmonary respiration; *C. diff*, *Clostridium difficile*.

fusion surgery for neuromuscular disease. Frailty is a complex term defined as a physiologic state of greater vulnerability to adverse health outcomes, and it was originally used to describe individuals in the geriatric population.[10,17] Frail patients comprise a significant proportion of the elderly population, but patients of all ages can exhibit dysregulation of multiple physiologic systems, increasing their risk of adverse outcomes after spinal fusion.[9,18] The Fried frailty phenotype was created to standardize the definition of frailty, distinguish it from comorbidity and disability, and pose a link between frailty and multisystem dysfunctions (ie, pulmonary dysfunction with low forced expiratory volume).[10,19] The Canada Study of Health and Aging Frailty Index (CSHA-FI), which included 70 measurements (eg, cognitive function, nutritional status), was created to evaluate these functions.[17] Although it effectively predicts surgical morbidity, collecting the numerous variables proved to be time consuming, which inspired more concise frailty measurements such as the mFI-11 and mFI-5.[20]

The mFI-11 was developed using NSQIP data and included 11 items from the CSHA-FI. However, its effectiveness has been questioned because of the frequently reported irregular recordings of the 11 variables.[20] This prompted the development of the mFI-5, which consists of just 5 variables: congestive heart failure, diabetes mellitus, chronic obstructive pulmonary disease or

Table 8. Univariate Logistic Regression and Frequency Analysis Depicting the Association of PV-5 Scores With Mortality.

PV-5 Score Cutoff	Or (95% CI)	Chi Squared	Cramer's Phi	*P* Value
1	2.14 (0.65-7.05)	1.63	0.013	0.2
2	2.28 (1.06-4.69)	4.7013	0.022	**0.030**
3	3.00 (1.40-6.43)	8.84	0.031	**0.003**
4	2.15 (0.29-15.90)	0.5921	0.080	0.442

P values in boldface type are significant at $P < 0.05$.
PV-5, 5-factor physiologic vulnerability score; OR, odds ratio; CI, confidence interval.

Table 9. Multivariate Logistic Regression Results of Prediction Model With Combined Odds Ratios and Probabilities.

Variables	Combined OR	Probability
CONMAL	1.30	5.29%
CNSAB	1.32	5.37%
HEMODIS	1.40	5.67%
CNSAB + CONMAL	1.72	6.87%
HEMODIS + CONMAL	1.82	7.25%
CNSAB + HEMODIS	1.85	7.36%
NUTRSUPP	2.21	8.67%
WNDINF	2.30	8.99%
CNSAB + HEMODIS + CONMAL	2.40	9.35%
NUTRSUPP + CONMAL	2.87	10.99%
CNSAB + NUTRSUPP	2.92	11.14%
WNDINF + CONMAL	2.99	11.39%
CNSAB + WNDINF	3.04	11.54%
NUTRSUPP + HEMODIS	3.09	11.73%
WNDINF + HEMODIS	3.22	12.15%
CNSAB + NUTRSUPP + CONMAL	3.79	14.00%
CNSAB + WNDINF + CONMAL	3.94	14.48%
NUTRSUPP + HEMODIS + CONMAL	4.02	14.73%
CNSAB + NUTRSUPP + HEMODIS	4.08	14.92%
WNDINF + HEMODIS + CONMAL	4.19	15.26%
CNSAB + WNDINF + HEMODIS	4.25	15.44%
WNDINF + NUTRSUPP	5.08	17.93%
CNSAB + NUTRSUPP + HEMODIS + CONMAL	5.31	18.58%
CNSAB + WNDINF + HEMODIS + CONMAL	5.53	19.20%
WNDINF + NUTRSUPP + CONMAL	6.61	22.12%
CNSAB + WNDINF + NUTRSUPP	6.71	22.38%
WNDINF + NUTRSUPP + HEMODIS	7.12	23.43%
CNSAB + WNDINF + NUTRSUPP + CONMAL	8.72	27.26%
WNDINF + NUTRSUPP + HEMODIS + CONMAL	9.25	28.44%
CNSAB + WNDINF + NUTRSUPP + HEMODIS	9.39	28.75%
CNSAB + WNDINF + NUTRSUPP + HEMODIS + CONMAL	12.21	34.41%
	Brier score loss	0.0906
	Accuracy	89%

OR, odds ratio; CONMAL, congenital malformation; CNSAB, CNS abnormality; HEMODIS, hematological disorder; WNDINF, preoperative wound infection; NUTRSUPP, preoperative nutritional support.

pneumonia, dependent functional health status, and hypertension requiring medication. Both mFI-5 and mFI-11 correlate frailty with morbidity and mortality in 9 different surgical specialties across various indications and case severities.[21] However, many of these measurements are not as relevant for the pediatric NMS population. The prevalence of many different frailty measurements underscores the need for an improved understanding of the physiologic basis of frailty and how it applies to various patient populations.[9]

The heterogeneity of various disease states limits the creation of a single generalizable and clinically relevant frailty metric.[9] Our results suggest that complications in pediatric patients with NMS are not associated with the same factors as those in adult patients in general but instead have a unique profile of factors contributing to postoperative outcomes. Within the context of adult spinal deformity, preoperative risk stratification tools such as the Charlson Comorbidity Index, the mFI-5, and the adult spinal deformity frailty index generate robust postoperative complication predictions.[22-29] However, these models were not created for children, including those with neuromuscular disorders. Recently developed risk stratification calculators, such as the BrAIST-Calc, have demonstrated significant prognostic benefits for the adolescent idiopathic scoliosis (AIS) population.[30] However, these calculators are geared more towards predicting curve progression than assessing surgical risk. Unlike the AIS population, children with NMS represent a high-risk surgical group, and every effort should be made to reduce the potential morbidity for posterior spinal fusion surgery.[3,31-34]

In our study, we found that postoperative complications were associated with 5 variables that can easily be obtained and used to stratify the risk for patients with NMS requiring deformity correction surgery. Furthermore, we have identified factors that beget vulnerability in this population, which is our key outcome of interest. The final PV-5 score includes congenital factors (malformations, structural CNS abnormalities), preoperative suggestions of physiologic impairment (nutritional support, previous wound infection), and hematologic conditions. Importantly, preoperative wound infection was defined as an open wound or an open wound infection present before surgery and did not refer to any procedure performed to address a surgical site infection. Interestingly, Jain et al.[35] reported similar factors (ie, seizures, nonverbal status, gastrostomy tube) that effectively predicted complications and health-related quality of life in patients with Gross Motor Function Classification System Level 5 cerebral palsy. These results align with previous studies that have reported higher complication rates for pediatric patients with NMS with similar preoperative comorbidities.[3,34,36-39] Furthermore, patients with PV-5 scores of 2 or greater demonstrated 2-3 times higher odds of mortality, although the relatively low frequency of mortality observed in our population must be

Table 10. Area Under the Curve Analysis to Evaluate PV-5 Model Performance Using the Validation Dataset.

	All Ages			Ages 0-9 years			Ages 10+ years		
PV-5 Score Cutoff	Accuracy	Sensitivity	Specificity	Accuracy	Sensitivity	Specificity	Accuracy	Sensitivity	Specificity
1	0.29	0.2	0.92	0.2	0.08	0.98	0.3	0.22	0.91
2	0.55	.054	0.67	0.4	0.34	0.78	0.59	0.57	0.66
3	0.7	0.74	0.44	0.59	0.6	0.53	0.72	0.76	0.44
4	0.79	0.88	0.19	0.73	0.79	0.37	0.81	0.9	0.22
5+	0.89	0.99	0.03	0.86	0.98	0.05	0.87	0.99	0.04

PV-5, 5-factor physiologic vulnerability score.

considered. These unique features imply a population-specific risk stratification profile for children with neuromuscular diseases requiring posterior spinal fusion that is different from that of adults undergoing spine surgery.[26-28,40-42] For example, cardiac factors (excluding congenital cardiac malformations)—which have been highly correlated with frailty status in adults—were found in 16% of patients in our study and did not significantly predict complications. This underscores the complexity of composite measures that may require more granular analyses of individual variables to interrogate relevant associations. Interestingly, the presence of hematologic disorder or preoperative wound infection (unrelated to previous NMS surgery) was rare, but these were significant contributors in our final model. These results suggest that patients with NMS exhibit a unique set of factors contributing to their vulnerability phenotype.

Age, when analyzed as a continuous variable, did not significantly influence complication rate; however, when categorizing patients by age, we found differences in the prognostic abilities of PV-5. The PV-5 score based on children of all ages significantly predicted complications at all levels for those ≥10 years old. Prognostication of complications in patients younger than this, however, was not effective with the PV-5 score. Limited sample size may have skewed these results, although it is possible that younger patients undergoing surgery (ie, before the end of growth potential) carry a different risk profile than older children. This may be due to a differential maturity in immune profile as children age that may influence the risk of infection and poor wound healing.[43] Additionally, differences in scoliosis severity may warrant earlier treatment, which is not reflected within the PV-5 and may account for its poor performance in this population. Such risk profiles may be associated with a different state of vulnerability and warrant further investigation, especially given the known high rate of complications with growth-friendly instrumentation in the early-onset scoliosis population.[10]

In children ≥10 years old, a PV-5 score of >3 accurately predicted complications for 74% of patients (accuracy = 70%, sensitivity = 74%, specificity = 44%). Patients with a PV-5 score of 5+ had complications nearly 100% of the time, although this accuracy is largely driven by sensitivity rather than specificity. Receiver operating curve analysis confirmed these findings, demonstrating our model's ability to effectively distinguish between true positives and true negatives, as evidenced by our high sensitivity and accuracy. It is important to note that our VIF results demonstrated minimal multicollinearity among variables in our multivariable model, indicative of the accuracy of our model, particularly with minimization of overfitting and the reduction of variance between our predictive variables.

After using PV-5 to guide our model development, we were able to compare the risk of developing a complication given every combination of comorbidities from PV-5. This allowed us to develop a risk calculator that uses binary inputs ("yes" or "no") to generate probabilistic risk outputs for every comorbidity. This model can enable earlier identification of patients with high-risk, physiologically vulnerable NMS, allowing clinicians to implement practices that can mitigate the incidence of postoperative complications. These include having appropriate discussions with family, deferring surgery, medically optimizing patients, diverting patients to higher-volume centers, and more appropriately allocating resource usage to support physiologically vulnerable patients. Patients with higher PV-5 scores may benefit from more postoperative follow-up and a lower threshold for intensive care unit admission or extended intensive care unit stay.

These broad clinical factors alone do not suggest frailty in the NMS population but rather serve as a preliminary analysis to better understand which body systems broadly predispose children to adverse events after surgery. This study builds upon the limited understanding of physiologic vulnerability within the pediatric population, a domain not well investigated using traditional frailty metrics. Physiologic vulnerability within the pediatric NMS population is composed of a distinct subset of comorbidities that may not directly align with those traditionally found in adults. We identified domains of interest that operationalize the vulnerable phenotype within pediatric patients with NMS, establishing a relationship between physiologic vulnerability and morbidity after deformity surgery. Future studies are warranted to interrogate more specific metrics within

Table 11. Predictive Risk Factor Definitions.

Variables	Definition
Congenital malformation	Patients with a history of a congenital defect at the time of surgery
Preoperative wound infection	Patients who have open wound/wound infection
Structural CNS abnormality	Patients who have any current or history of a structural CNS abnormality documented in the medical record: Myelomeningocele; microcephaly; macrocephaly; anencephalus; congenital hydrocephalus; hypotelorism; trigonocephaly; congenital Chiari malformation; Dandy-Walker malformation; Arnold-Chiari malformation; syrinx; tethered cord syndrome; grey/white matter changes agenesis of the corpus callosum; fused ventricles; aqueductal stenosis; other neural tube defects; other cystic or degenerative lesions; encephalocele; spina bifida; holoprosencephaly. These may also be noted in a visual or radiologic examination
Preoperative nutritional support	Patients who are receiving nutritional support at the time of the primary procedure: Enteral feedings via orogastric, nasogastric, gastrostomy, nasojejunal, or jejunostomy tubes; intravenous TPN.
Hematologic disorder	Patients who have disorders affecting the hematologic system (blood), which includes plasma, platelets, leukocytes, and erythrocytes and the bone marrow: Sickle cell disease; thalassemia; hereditary spherocytosis; thrombocytopenia; idiopathic thrombocytopenic purpura; heparin-induced thrombocytopenia; neutropenia; Henoch-Schoenlein disease; hemolytic, hypoproliferative, macrocytic, microcytic, normocytic, pernicious, or Fanconi's/aplastic anemia; basophilia; dysfibrinogenemia; eosinophilia; vitamin K deficiency; hemophilia; platelet function defects; von Willebrand's disease; congenital antithrombin III deficiency; congenital protein C or S deficiency; disseminated intravascular coagulation; factor II, V, VII, X, or XII deficiencies

CNS, central nervous system; TPN, total parenteral nutrition.

physiologic systems to potentially enhance the predictive capabilities of PV-5 and other measures of physiologic vulnerability in pediatric patients.

Limitations

The retrospective design of this study limits the ability to draw definitive conclusions regarding the prognostic abilities of the PV-5. Additionally, the use of the NSQIP database, although it is robust and well documented, may limit the generalizability and performance of the PV-5 score in other populations; however, most frailty studies are retrospective database studies with similar limitations.[21,40,44,45] The NSQIP data also have limited follow-up times, preventing the analysis of long-term complications. Further, because variables to be considered were identified by a group consensus, there is a small risk of selection bias; we aimed to reduce this by including nearly all preoperative variables.

It is important to note that we only looked at patients with NMS who underwent spinal fusion. These patients undergo a variety of surgeries of varying complexities, which may include risk factors and complications distinct from those found in our heterogeneous sample. We did not stratify complications by type or severity but did choose to report the incidence of each. Although this may limit the granularity of the PV-5, the goal of this study was to robustly report associations with any complication and use this as a proxy for morbidity. Specific system-based complications would be of interest but are outside the scope of this exploratory analysis; however, the complications chosen in the composite profile are well-established, tracked metrics and thus justifiable within our chosen methodology. Lastly, we relied on reported CPT and ICD-9 and -10 codes to refine our patient population. This may lead to an underreporting of the number of patients with NMS and their potential risk of postoperative complications.

Although these limitations must be considered, there are very few studies that attempt to define physiologic vulnerability within the pediatric population. This study is the first to develop and implement a score to predict postoperative complications in patients with NMS undergoing deformity correction surgery. The PV-5 score can be used as an effective risk-stratification tool to optimize patient selection preoperatively and reduce adverse postoperative complications. This study is an important pilot for future studies that validate the predictive capabilities of this scoring system and expand it into other pediatric disease contexts. Future study will require further examination of the specific conditions encapsulated in each of the broad categories captured in the PV-5 score (Table 11). Such analysis may uncover additional vulnerability characteristics that can improve prognostication within the pediatric NMS population.

Conclusion

Physiologic vulnerability in children with neuromuscular disease, as indicated by higher PV-5 scores, was associated with significantly greater incidence of complications after pediatric NMS deformity correction surgery. The PV-5 score and prediction model can be used as an effective risk-stratification tool to help surgeons optimize patients preoperatively and reduce adverse postoperative complications. In the future, task-based variables applicable to the

NMS population will be useful to further enhance the definition of physiologic vulnerability in this population.

Abbreviations

NMS	neuromuscular scoliosis
AUC	area under the curve
OR	odds ratio
PV-5	5-factor physiologic vulnerability score
mFI	modified frailty index
NSQIP-P	National Surgical Quality Improvement Program Pediatric database
ICD	International Classification of Diseases
CPT	Current Procedural Terminology
SSI	surgical site infection
AIC	Akaike information criterion
CI	confidence interval
ASA	American Society of Anesthesiologists
CNS	central nervous system
CSHA-FI	Canada Study of Health and Aging Frailty Index
VIF	variance inflation factor

Author Contributions

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Declaration of Conflicting Interests

The author(s) declared the following potential conflicts of interest with respect to the research, authorship, and/or publication of this article: Jennifer Bauer is a consultant for Orthopaediatrics, Depuy Synthes, and Proprio; serves on the Board of Directors of the Pediatric Orthopaedic Society of North America; and her institution has received a quality improvement grant from POSNA. Christopher P. Ames is a consultant for DePuySynthes, Medtronic, Medicrea, K2M, Agada Medical, and Carlsmed; receives royalties from Stryker, Biomet Zimmer Spine, DePuy Synthes, Nuvasive, Next Orthosurgical, Medicrea, and K2M; has research support from Titan Spine, DePuy Synthes, and ISSG; and has a grant from the Scoliosis Research Society. Peter O. Newton is a consultant for DePuy Synthes Spine, Stryker/K2M, MiRus, Globus Medical, and Pacira; has received honoraria from DePuy Synthes Spine and Medtronic; receives royalties from DePuy Synthes Spine, Thieme Publishing, and Stryker/K2M; has patents with DePuy Synthes Spine (anchoring systems and methods for correcting spinal deformities [8540754], low-profile spinal tethering systems [8123749], screw placement guide [79811117], compressor for use in minimally invasive surgery [7189244]) and Stryker/K2M (posterior spinal fixation); is on the leadership of the Scoliosis Research Society, the Setting Scoliosis Straight Foundation, the Rady Children's Specialists of San Diego, and the International Pediatric Orthopedic Think Tank; has equity interest in Accelus and Spinology; and his institution receives grant funding from DePuy Synthes Spine, the Scoliosis Research Society, EOS Imaging, Nuvasive, Orthopediatrics, Stryker/K2M, Alphatech, Mazor Robotics, and the Setting Scoliosis Straight Foundation. The other authors report no conflicts of interest.

Ethical Statement

Ethical Approval

Because NSQIP-P participant user file datasets do not have patient identifying information, the study was exempt from ethics review as determined by the local institutional review board and human subjects' protection office.

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Data Availability Statement

Data are available upon reasonable request to the corresponding author.

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